

Signal Processing

Exercise Sheet 7

Summer Semester 2026

Part 1 – Laplace Transform

Please be informed that we expect you to be already familiar with the Laplace Transform. If this is not the case, please make sure you know the basic properties of the Laplace Transform, how stability can be determined with help of the transfer function in the Laplace domain and how to sketch and read a block diagram by the time you take the exam.

Definition of the Laplace Transform

$$F(s) = \int_0^{\infty} f(t) \cdot e^{-st} dt,$$

in case $f(t) = 0$ for $t < 0$.

Possible notations

$$F(s) = \mathcal{L}\{f(t)\}$$

$$f(t) \circ \longrightarrow \bullet F(s)$$

Inverse Laplace Transform

$$\mathcal{L}^{-1}\{F(s)\} = f(t)$$

$$F(s) \bullet \longrightarrow \circ f(t)$$

Properties of the Laplace Transform

You are supposed to be familiar with following properties:

- | | | |
|----------------|-------------------|-----------------------|
| - Linearity | - Time Shifting | - Frequency Shifting |
| - Time Scaling | - Differentiation | - Integral Properties |
| - Convolution | | |

Question 1 – Laplace Transform

Figure 1 shows a simple version of a low-pass filter.

- Determine the corresponding differential equation in both the time and Laplace domain. For the sake of simplicity, we will assume $u_o(t = 0 \text{ s})$ to be 0 V.
- Determine the step response in both the time and Laplace domain. Please use the partial fraction decomposition to transform your equation from the Laplace domain to the time domain.
- Decide on whether the system is stable or unstable.

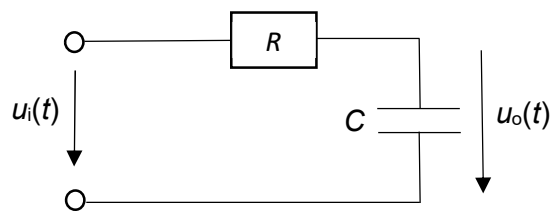


Figure 1.: Low-pass filter.

Solution – 1a)

$$u_i(t) = i(t) \cdot R + u_o(t) \rightarrow U_o(s) = \frac{U_i(s)}{RCs + 1}$$

Solution – 1b)

Heavyside Function: $U_i(s) = \frac{1}{s} U_E$, $U_E = \text{const.} = 1$

$$U_o(s) = \frac{U_E}{s} - \frac{U_E}{s + \frac{1}{RC}}$$

$$u_o(t) = U_E \left(\sigma(t) - e^{\frac{-t}{RC}} \right) = U_E \left(1 - e^{\frac{-t}{RC}} \right) = 1 - e^{\frac{-t}{RC}}$$

Solution – 1c)

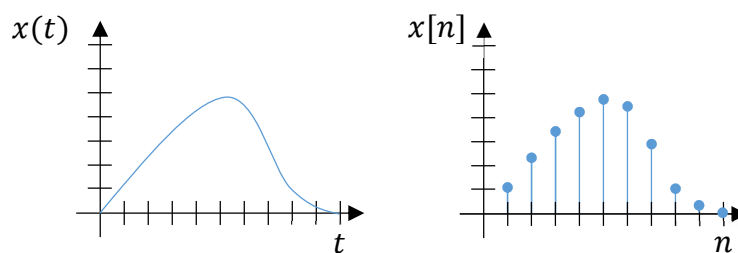
$$RCs + 1 = 0$$

$$s = -\frac{1}{RC}$$

The system is stable if we assume $RC > 0$.

Part 2 – Discrete Time (DT) Signals and Z-Transform

Sampling and quantization of input signals allows for the analysis of the conveyed information in the received signal with the help of microcontrollers. However, this also implies that values of the sampled signal are only known at certain points of time, which can be described by numerical sequences $x[n]$.



Since the sampled signal is both discrete in time and value, mathematical operations such as integration and differentiation cannot be applied. It is for this reason, that we will describe systems with the help of difference equations instead. These are the discrete equivalents to differential equations.

Fill in the gaps to determine the difference equation of the given system in Figure 2.

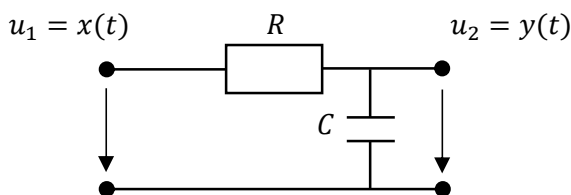


Figure 2.: Low-pass filter.

$$(I) \quad i_C = i_R \rightarrow C \cdot \frac{du_2(t)}{dt} = \frac{u_1(t) - u_2(t)}{R}$$

(II) $u_i(t) \rightarrow u_i[n]$; Keep in mind: A difference quotient (DQ) for discrete time systems is the equivalent of a differentiation for continuous time systems.

Mind: $u[n] = u(kT_s)$

DQ: $\frac{du_2(t)}{dt} \approx \frac{u_2[n] - u_2[n-1]}{T_s}$

$$C \cdot \frac{u_2[n] - u_2[n-1]}{T_s} = \frac{u_1[n] - u_2[n]}{R} .$$

Solving for $u_2[kT_s]$ results in

$$u_2[kT_s] = \underbrace{\frac{T_s}{RC+T_s}}_b \cdot u_1[n] + \underbrace{\frac{RC}{RC+T_s}}_a u_2[n-1] .$$

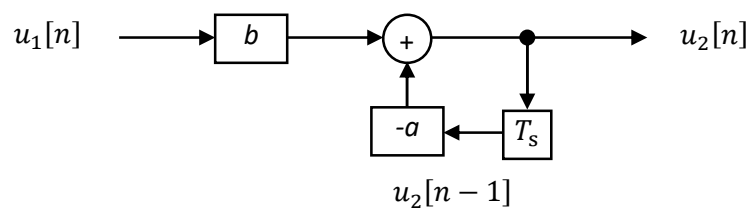
Difference Equation:

$$y[n] = \sum_{i=0}^N b_i \cdot x[n-i] - \sum_{i=1}^M a_i \cdot y[n-i]$$

→ “Standardized” difference equation

$i = 1$, since we assume that only current or past values impact our output signal $y[n]$.

→ block diagram of the “standardized” difference equation



Properties of discrete time systems

Linearity

A system is referred to as linear if each linear combination of input signals $x_1[n]$, $x_2[n]$... and arbitrary constants v_1 , v_2 ..., e.g.,

$$x[n] = v_1 \cdot x_1[n] + v_2 \cdot x_2[n]$$

leads to the respective linear combination of output signals, in our case

$$y[n] = v_1 \cdot y_1[n] + v_2 \cdot y_2[n].$$

This implies that both principles of homogeneity and superposition hold.

Time-Invariance

A discrete time system is referred to as time-invariant if the system response $y_r(n)$ is delayed to the same extent as the input signal $x(n)$.

$$\sum_{i=0}^N b_i \cdot x[n - i - n_0] = \sum_{i=0}^M a_i \cdot y_r[n - i - n_0]$$

Causality/Causation

A system is referred to as causal if its output only depends on current or past values of its input.

In contrast, the output of an anticausal system depends only on future input values, the output of a non-causal system usually depends on both past and future input values.

Example of a system that is causal: $y[n] = 2(x[n - 1] - 2x[n] - y[n - 1])$

Stability

A discrete time system is stable if the region of convergence (ROC) contains the unit circle.

Only if the system is **causal**, the system is stable in case all poles of the transfer function are inside the unit circle.

The Z-transform

Difference equations of discrete time systems are the equivalent of the differential equations of time continuous systems.

In fact, one can consider the Z-transform to be the counterpart of the Laplace-transform for discrete time systems.

Properties of the Z-transform

	Sequence	Z-transform
Linearity	$\alpha \cdot x_1[n] + \beta \cdot x_2[n]$	$\alpha \cdot X_1(z) + \beta \cdot X_2(z)$
Time Shifting to the right	$x[n - i]$ for $i > 0$	$z^{-i} \cdot X(z)$
Time Shifting to the left	$x[n + i]$ for $i > 0$	$z^i \cdot X(z) - z^i \cdot \sum_{m=0}^{i-1} x[m] \cdot z^{-m}$
Damping	$a^n \cdot x[n]$	$X\left(\frac{z}{a}\right)$
Linear Weighting	$n \cdot x[n]$	$-z \cdot \frac{d}{dz} \cdot X(z)$
Differentiation	$x[n] - x[n - 1]$	$\frac{z - 1}{z} \cdot X(z)$
Sum	$\sum_{n=0}^N x[n]$	$\frac{z}{z - 1} \cdot X(z)$
Convolution	$x_1[n] * x_2[n]$	$X_1(z) \cdot X_2(z)$

The Z-transform often simplifies solving for a difference equation of a system.

$$y[n] = \sum_{i=0}^N b_i x[n - i] - \sum_{i=1}^M a_i y[n - i] \quad Y(z) = \sum_{i=0}^N b_i X(z) \cdot z^{-i} - \sum_{i=1}^M a_i Y(z) \cdot z^{-i}$$

Transfer function/system function/impulse response $H(z)$

$$Y(z) + \sum_{i=1}^M a_i Y(z) \cdot z^{-i} = \sum_{i=0}^N b_i X(z) \cdot z^{-i}$$

$$Y(z) \cdot \left(1 + \sum_{i=1}^M a_i \cdot z^{-i}\right) = X(z) \sum_{i=0}^N b_i \cdot z^{-i}$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{i=0}^N b_i \cdot z^{-i}}{1 + \sum_{i=1}^M a_i \cdot z^{-i}} = \frac{b_0 + b_1 \cdot z^{-1} + b_2 \cdot z^{-2} + \dots + b_N \cdot z^{-N}}{1 + a_1 \cdot z^{-1} + a_2 \cdot z^{-2} + \dots + a_M \cdot z^{-M}}$$

Number of **zeros**: polynomial degree N of the **numerator**

zeros: numerator polynomial = 0

Number of **poles**: polynomial degree M of the **denominator**

poles: denominator polynomial = 0

Z-Transform of causal sequences $x[n]$, $n \geq 0$:

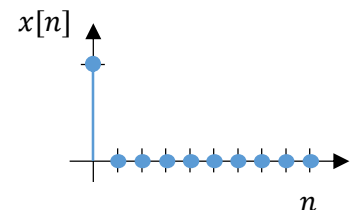
$$X(z) = \sum_{n=0}^{\infty} x[n] \cdot z^{-n}.$$

Elementary signals and their corresponding z-transform

Dirac Delta Impulse $x[n] = \delta[n]$

$$X(z) = \sum_{n=0}^{\infty} \delta[n] \cdot z^{-n}$$

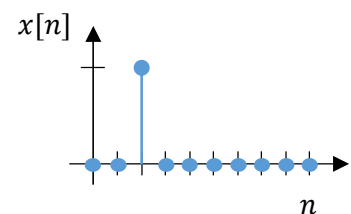
$$X(z) = z^{-0} = 1$$



Delayed Dirac Delta Impulse $x[n] = \delta[n - M]$

$$X(z) = \sum_{n=0}^{\infty} \delta[n - M] \cdot z^{-n} = z^{-M}$$

$$X(z) = z^{-M}$$



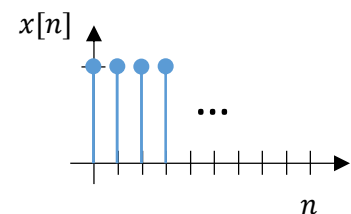
Heaviside Function $x[n] = \sigma[n]$

$$X(z) = \sum_{n=0}^{\infty} \sigma[n] \cdot z^{-n} = \sum_{n=0}^{\infty} z^{-n} = \sum_{n=0}^{\infty} (z^{-1})^n$$

Geometric sum for infinite terms

$$\sum_{n=0}^{\infty} q^n = \frac{1}{1-q}$$

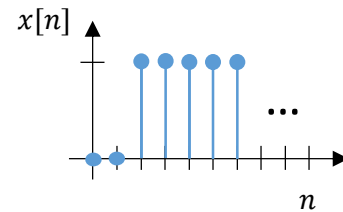
$$X(z) = \frac{z}{z-1}$$



Delayed Heaviside Function $x[n] = \sigma[n - M]$

$$X(z) = \sum_{n=0}^{\infty} \sigma[n - M] \cdot z^{-n}$$

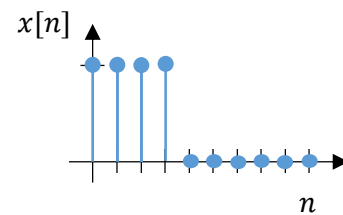
$$X(z) = \frac{z}{z - 1} \cdot z^{-M}$$



Finite Unit Step Function

$$x[n] = \sigma[n] - \sigma[n - M]$$

$$X(z) = \sum_{n=0}^{\infty} (\sigma[n] - \sigma[n - M]) \cdot z^{-n} = \sum_{n=0}^{M-1} z^{-n}$$



Geometric sum for finite terms: $\sum_{n=0}^{M-1} q^n = \frac{1-q^M}{1-q}$

$$X(z) = \frac{1 - z^{-M}}{1 - z^{-1}}$$

Question 1

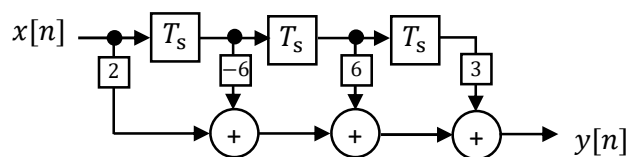


Figure 3.: Block diagram of a discrete system.

Figure 3 shows a block diagram of a discrete system.

- Determine the difference equation of the system.
- Determine the transfer function $H(z)$.
- Determine the zeros and poles of the transfer function $H(z)$.
- Determine the impulse response $h(n)$.

Solution – 1a)

$$y[n] = 2 \cdot x[n] - 6 \cdot x[n-1] + 6 \cdot x[n-2] + 3 \cdot x[n-3]$$

Solution – 1b)

$$Y(z) = 2 \cdot X(z) \cdot z^0 - 6 \cdot X(z) \cdot z^{-1} + 6 \cdot X(z) \cdot z^{-2} + 3 \cdot X(z) \cdot z^{-3}$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{2 - 6z^{-1} + 6z^{-2} + 3z^{-3}}{1}$$

$$H(z) \cdot \frac{z^3}{z^3} = \frac{2z^3 - 6z^2 + 6z + 3}{z^3}$$

Solution - 1c)

$$\text{Zeros: } z_{0,k}: 2z^3 - 6z^2 + 6z + 3 = 0$$

$$\rightarrow z_{0,1} = 1,68 + 1,18j; z_{0,2} = 1,68 - 1,18j; z_{0,3} = -0,36$$

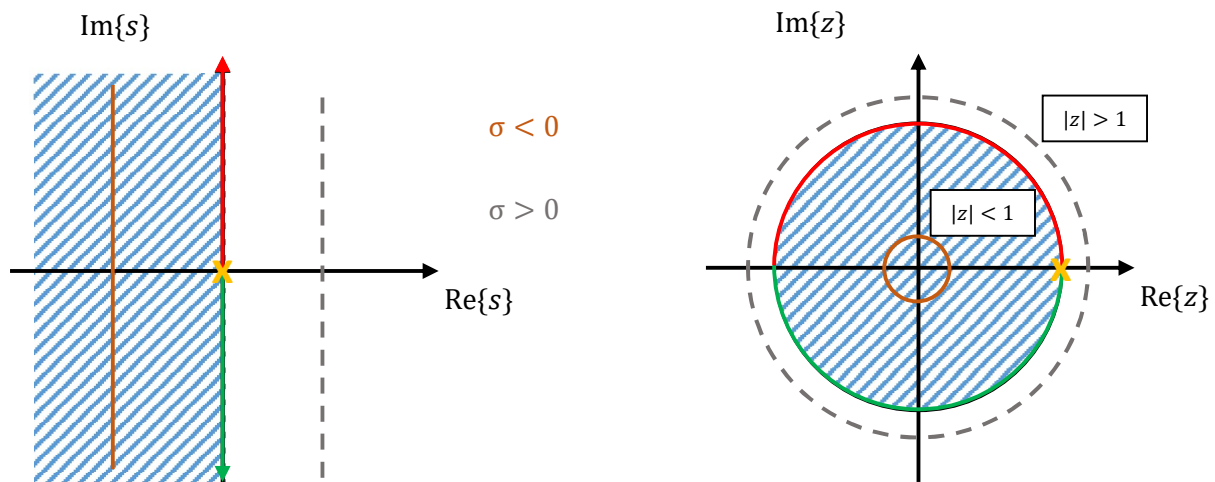
$$\text{Poles: } z_{\infty,k}: z^3 = 0$$

$$\rightarrow z_{\infty,1-3} = 0$$

Solution – 1d)

$$h[n] = 2\delta[n] - 6\delta[n-1] + 6\delta[n-2] + 3\delta[n-3]$$

Relationship between the z-plane and s-plane



Keep in mind: $z = e^{sT_s}$

Question 2

Sketch *approximately* to each point in the s-plane (Figure 4) a corresponding point in the z-plane (Figure 5).

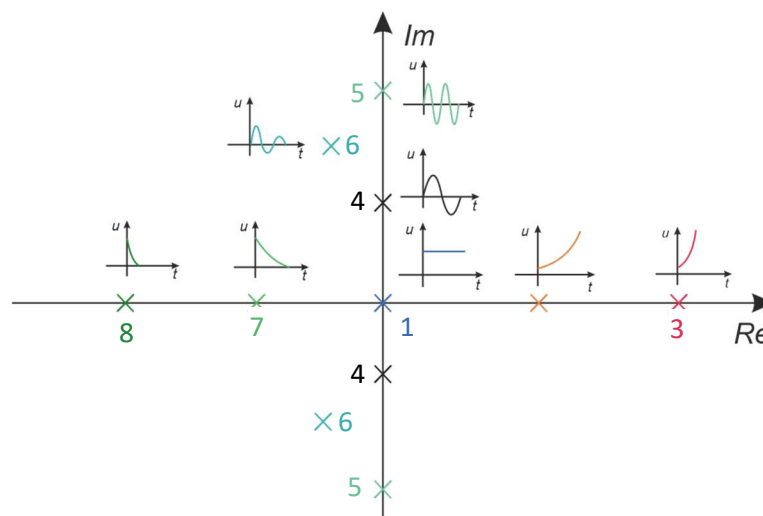


Figure 4.: s-plane.

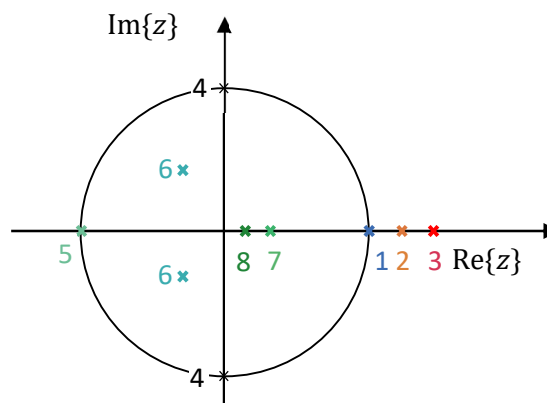


Figure 5.: z-plane.